Online Appendix: Admission channels and origin diversification in Polish labour migration from 2008 to 2025

Supplementary tables A1-A14. Journal of Ethnic and Migration Studies. Data and code: Zenodo, doi:10.5281/zenodo.[XXXXXXX] (DOI to be added at acceptance).

The unit of observation is an administrative document, not a unique person. All data are aggregated by citizenship. Table numbers in the article (Tables 1-6, Figures 1-5) refer to the main text.

DRAFT. Tables marked 'draft' contain the values reported in the article; cells shown in red ([run], [confirm], [page], [authors]) are produced by the replication code or must be supplied by the authors before submission. Regenerate this file with code/run_all.py once the data are in place.

Generated 2026-11-02 by code/11_build_appendix.py.

Contents

Table A1. Source files [draft]
Table A2. Annual documents by channel, 2008-2025 (data of Figure 1) [draft]
Table A3. Concentration series: effective number of origins (1/HHI) (data of Figure 3) [draft]
Table A4. Origins and estimation samples [draft]
Table A5. Policy coding: declaration access, EU visa freedom and shock indicators [draft]
Table A6. Group difference in persistence: alternative inference [draft]
Table A7. Additional persistence estimates by access group [draft]
Table A8. Changes in declaration access: DiD summaries, event-study coefficients and placebo distributions [draft]
Table A9. Interrupted time series with standard errors [draft]
Table A10. Duration-sensitive concentration [draft]
Table A11. Applications and issuance, 2025 [draft]
Table A12. Person-based benchmarks [draft]
Table A13. Variable definitions [draft]
Table A14. Software and runtime [draft]

Table A1. Source files

Series	Period	Frequency	Files	Portal	Accessed	Revision note
Work permits by citizenship	2008-2025	annual	[run]	MRPiPS (n.d.)	2026-10-25	revised 2021 tables published in 2025 are used
Work permits by citizenship	2008H1-2025H2 (36 half-years)	half-yearly	[run]	MRPiPS (n.d.)	2026-10-25	
Work permits by citizenship	2021-2025	monthly	[run]	MRPiPS (n.d.)	2026-10-25	
Employer declarations by citizenship	2007-2025	monthly	[run]	MRPiPS (n.d.)	2026-10-25	entered into register from 2018
Persons covered by declarations	2019-2025	annual	[run]	MRPiPS (n.d.)	2026-10-25	
Seasonal work permits by citizenship	2018-2025	monthly	[run]	MRPiPS (n.d.)	2026-10-25	
Notifications (Ukrainian citizens)	2022-2025	monthly	[run]	MRPiPS (n.d.)	2026-10-25	March 2022 is a partial month
Total			88			

Notes: One row per spreadsheet downloaded from the statistics pages of the Public Employment Services portal (MRPiPS n.d.). The full inventory with URL, download date and SHA-256 checksum is data/raw/mrpips/source_inventory.csv. Where MRPiPS revised a table (the 2021 tables republished in 2025), the superseded file is kept with use = 0 and is not read.

Source: MRPiPS (n.d.), accessed 25 October 2026.

Status: draft.

Table A2. Annual documents by channel, 2008-2025 (data of Figure 1)

Year	Work permits	Declarations	Seasonal permits	Notifications	Total documents	Insured foreigners (ZUS, 31 Dec)
2008	[run]	[run]	NA	NA	175,000	[report]
2009	[run]	[run]	NA	NA	[run]	[report]
2010	[run]	[run]	NA	NA	[run]	[report]
2011	[run]	[run]	NA	NA	[run]	[report]
2012	[run]	[run]	NA	NA	[run]	[report]
2013	[run]	[run]	NA	NA	[run]	101,000
2014	[run]	[run]	NA	NA	[run]	[run]
2015	[run]	[run]	NA	NA	[run]	[run]
2016	[run]	[run]	NA	NA	[run]	[run]
2017	[run]	[run]	NA	NA	[run]	[run]
2018	[run]	[run]	134,600	NA	[run]	[run]
2019	[run]	[run]	[run]	NA	[run]	[run]
2020	[run]	[run]	[run]	NA	[run]	[run]
2021	[run]	[run]	[run]	NA	2,600,000 (peak)	[run]
2022	[run]	[run]	[run]	[run]	[run]	[run]
2023	[run]	[run]	[run]	[run]	[run]	[run]
2024	[run]	[run]	[run]	[run]	[run]	[run]
2025	[run]	[run]	[run]	[run]	1,630,000	1,289,000

Notes: Documents issued or registered during the year; not unique persons. NA: channel not yet in operation (seasonal permits before 2018, notifications before 2022). ZUS: stock of insured foreigners at the end of the year, as reported in MRPiPS (2026).

Source: Authors' calculations based on MRPiPS registers; ZUS stock as reported in MRPiPS (2026).

Status: draft.

Table A3. Concentration series: effective number of origins (1/HHI) (data of Figure 3)

Period	Work permits	All documents excl. notifications	Excl. notifications, declaration persons	All documents
2008	7.9	[run]	NA	[run]
2009	[run]	[run]	NA	[run]
2010	[run]	[run]	NA	[run]
2011	[run]	[run]	NA	[run]
2012	[run]	[run]	NA	[run]
2013	[run]	[run]	NA	[run]
2014	[run]	[run]	NA	[run]
2015	[run]	[run]	NA	[run]
2016	1.4	[run]	NA	[run]
2017	[run]	[run]	NA	[run]
2018	[run]	[run]	NA	[run]
2019	[run]	[run]	[run]	[run]
2020	[run]	[run]	[run]	[run]
2021	2.4	[run]	[run]	1.6
2022	[run]	[run]	[run]	[run]
2023	[run]	[run]	[run]	[run]
2024	[run]	8.6	[run]	[run]
2025	13.9	8.6	9.5	1.9
Half-yearly work permits, 2008H1-2025H2	[run] (36 values)			

Notes: HHI = sum of squared origin shares of documents. Annual totals except the half-yearly work-permit series (36 half-years). 'Declaration persons' replaces the number of declarations by the number of persons covered (available from 2019).

Source: Authors' calculations based on MRPiPS registers.

Status: draft.

Table A4. Origins and estimation samples

Summary

Sample	Origins	Observations	Used in	Definition
Country-year frame	164	2,952 (164 x 18)	Figures 1-3; Table 2	all citizenships recorded in at least one series 2008-2025
Pooled model	148	2,150	Table 3 col. 1	2011-2025; non-missing covariates incl. distance
Fixed-effects baseline	148	2,184	Table 3 cols. 2-4	2011-2025; World Bank growth data; not only zero permits
Excluding Ukraine and Belarus	146	2,157	Table 3 col. 5	baseline without UKR and BLR
Never-eligible origins	142	2,097	Table 3 col. 6	baseline origins that never had declaration access
Pre-sample mean model	[run]	2,190	Table 4	all-zero origins retained when origin effects are dropped
Comparison origins	70	[run]	Table 5; Figure 4	never eligible and at least 500 documents 2008-2025
Full list of origins with exclusion reasons	164			[run] (output/tables/appendix_A4.csv)

Notes: Sample sizes refer to 2011-2025 (15 years). Origins without World Bank real GDP-per-capita growth data or with only zero outcomes are dropped from the fixed-effects models; all-zero origins are retained when origin effects are replaced by the pre-sample mean (Table 4). Comparison origins (Section 5.3): never eligible for declarations and at least 500 documents in 2008-2025.

Source: Authors' calculations based on MRPiPS registers and World Bank (2026).

Status: draft.

Table A5. Policy coding: declaration access, EU visa freedom and shock indicators

Origin	Declaration access	Legal basis	Partial-year access share	EU visa freedom from	Partial-year visa share	Shock indicators	Group indicators
Belarus	from 2006-09	Dz.U. 2006 nr 156 poz. 1116	-	-	-	Belarus 2021+	2018+; 2022+
Russia	2006-09 to 2022-10	Dz.U. 2006 nr 156 poz. 1116; removed by Dz.U. 2022 poz. 2204	2022: 0.830	-	-	Russia 2022+	2018+; 2022+
Ukraine	from 2006-09	Dz.U. 2006 nr 156 poz. 1116	-	2017-06-11	2017: 0.559	Ukraine 2014+; Ukraine 2022+	2018+; 2022+
Moldova	from 2009-02	Dz.U. 2009 nr 21 poz. 114	2009: 0.915	2014-04-28	2014: 0.679	-	2018+; 2022+
Georgia	2010-12 to 2025-12	Dz.U. 2010 nr 236 poz. 1559; removed by Dz.U. 2025 poz. 1617/1622	2010: 0.085; 2025: 0.915	2017-03-28	2017: 0.764	-	2018+; 2022+
Armenia	from 2014-01	Dz.U. 2013 poz. 1507	-	-	-	-	2018+; 2022+

Legal acts

act_id	dz_u	date_signed	date_in_force	date_precision	provision	effect	used_in	verify_note
A01	Dz.U. 2006 nr 156 poz. 1116	NA	2006-09	month	regulation on work without a permit (employer declarations)	declarations introduced for citizens of states bordering Poland (BY, RU, UA); agriculture only; 3 of 6 months	Table 1; access_spells	enter exact dates
A02	Dz.U. 2007 nr 120 poz. 824	NA	2007-07	month	amending regulation	declarations opened to all sectors	Table 1	enter exact dates
A03	Dz.U. 2008 nr 17 poz. 106	NA	2008-02	month	amending regulation	maximum period extended from 3 to 6 months within 12 months	Table 1; document_duration	enter exact dates
A04	Dz.U. 2009 nr 21 poz. 114	NA	2009-02	month	amending regulation	declarations extended to Moldova	access_spells (MDA)	enter exact dates
A05	Dz.U. 2010 nr 236 poz. 1559	NA	2010-12	month	amending regulation	declarations extended to Georgia	access_spells (GEO)	enter exact dates
A06	Dz.U. 2013 poz. 1507	NA	2014-01	month	amending regulation	declarations extended to Armenia	access_spells (ARM)	enter exact dates
A07	Dz.U. 2017 poz. 1543	NA	2018-01-01	day	amendment of the Act on employment promotion and labour market institutions (implementation of Directive 2014/36/EU)	register of declarations kept by county offices (refusal possible); declarations restricted to non-seasonal work; seasonal work permit; fees	Table 1; ITS 2018 break	enter date signed
A08	Dz.U. 2017 poz. 2349	NA	2018-01-01	day	implementing regulation (2018 reform)	list of states / procedural rules for declarations	Table 1	confirm content and dates
A09	Dz.U. 2017 poz. 2350	NA	2018-01-01	day	implementing regulation (2018 reform)	implementing rules (2018 reform)	Legal acts cited	confirm content and dates
A10	Dz.U. 2022 poz. 91	NA	2022-01-29	day	amendment	maximum period of work on a declaration raised to 24 months	Table 1; document_duration	enter date signed
A11	Dz.U. 2022 poz. 583	2022-03-12	NA	day	Act on assistance to citizens of Ukraine, art. 22	notification within 14 days lets any Ukrainian legally in Poland work; free; no decision	Table 1; ITS 2022 break	enter in-force date and retroactive effect
A12	Dz.U. 2022 poz. 2204	NA	2022-10	month	amending regulation	Russia removed from declaration countries; Belarus retained	access_spells (RUS)	enter exact in-force day
A13	Dz.U. 2024 poz. 854	NA	2024-07	month	amendment	notification deadline shortened from 14 to 7	Section 2	enter exact dates

act_id	dz_u	date_signed	date_in_force	date_precision	provision	effect	used_in	verify_note
						days		
A14	Dz.U. 2025 poz. 621	2025-03-20	2025-06-01	day	Act on the conditions for admitting foreigners to work in Poland	labour market test abolished (county lists of excluded occupations); online procedures; contract filing; refusal grounds; fines PLN 3000-50000; quotas	Table 1; ITS 2025 break	
A15	Dz.U. 2025 poz. 1617	NA	2025-12-01	day	implementing regulation	Georgia removed from declaration countries	access_spells (GEO)	confirm which of 1617/1622 removes Georgia and which sets fees
A16	Dz.U. 2025 poz. 1622	NA	2025-12-01	day	implementing regulation	fees raised	Section 2	see A15
A17	Dz.U. 2026 poz. 203	2026-01-23	2026-03-05	day	amendment of the Act on assistance to citizens of Ukraine	new notifications only for holders of temporary-protection (UKR) status; outside the 2008-2025 series	Table 1 (note)	

Notes: Access shares are the share of calendar days in the year on which the right applied (data/policy/access_spells.csv). EU visa freedom applies to holders of biometric passports. Group indicators equal one for the six declaration nationalities (ARM, BLR, GEO, MDA, RUS, UKR) from 2018 and from 2022.

Source: Authors' compilation based on the legal acts listed (Dziennik Ustaw) and European Commission (2026).

Status: draft.

Table A6. Group difference in persistence: alternative inference

Method	Difference (access minus other)	95% interval	p-value (H0: difference = 0)	Draws	Note
Point estimate (Table 3 col. 3: 0.661 - 0.652)	0.009				
Analytic cluster-robust	0.009	[run]	[run]		G/(G-1) correction
Pairs cluster bootstrap (submitted version)	0.009	-0.56 to 0.34		199	superseded; reported in the submitted manuscript
Pairs cluster bootstrap	0.009	[run]	[run]	999	percentile interval; unidentified draws reported
Wild cluster score bootstrap (Rademacher)	0.009	[run]	[run]	9999	null imposed; interval by test inversion
Wild cluster score bootstrap (Webb)	0.009	[run]	[run]	9999	null imposed; interval by test inversion

Notes: Difference between the recent-flow elasticities of origins with and without declaration access (Table 3, column 3), estimated as a single coefficient. Only six origins ever have access, so all intervals are approximate. Pairs bootstrap: origins resampled with replacement; draws that contain no access origin do not identify the difference and are reported. Wild cluster score bootstrap: Kline and Santos (2012), null imposed, Rademacher and Webb weights; the interval is the set of null values not rejected at 5 percent.

Source: Authors' estimates based on MRPiPS registers and World Bank (2026).

Status: draft.

Table A7. Additional persistence estimates by access group

Specification	Declaration-access origins	Other origins	Difference	Observations
Baseline (Table 3 col. 3)	0.661*** (0.113)	0.652*** (0.080)	[run]	2,184
Lagged counts from permits only	0.68 [run: s.e.]	0.68 [run: s.e.]	[run]	2,184
Excluding Ukraine	0.66 [run: s.e.]	0.68 [run: s.e.]	[run]	[run]
Without reform indicators, 2011-2024	[run]	[run]	[run] ('larger contrast' in Section 6.2)	[run]
Without reform indicators, 2011-2025	[run]	[run]	[run]	2,184
With reform indicators, 2011-2024	[run]	[run]	[run]	2,046
Leave one of the 20 largest origins out (Table 3 col. 2)	range 0.61 to 0.72		20 rows: [run]	

*Notes: PPML with origin and year effects; standard errors clustered by origin in parentheses. Controls as in Table 3, column 3, unless stated. *** p<0.01, ** p<0.05, * p<0.1.*

Source: Authors' estimates based on MRPiPS registers and World Bank (2026).

Status: draft.

Table A8. Changes in declaration access: DiD summaries, event-study coefficients and placebo distributions

DiD summaries and permutation p-values

Change	Outcome	Effect	p (k/N) as reported	k	N placebo	p (k+1)/(N+1)
Moldova enters (2009)	Work permits	-69%	0.20	[run]	[run]	[run]
Moldova enters (2009)	All documents excl. notifications	x3.6	0.19	[run]	[run]	[run]
Georgia enters (event year 0 = 2011)	Work permits	+64%	0.43	[run]	[run]	[run]
Georgia enters (event year 0 = 2011)	All documents excl. notifications	x7.8	0.01	[run]	[run]	[run]
Georgia enters (event year 0 = 2010)	Work permits	[run]	[run]	[run]	[run]	[run]
Georgia enters (event year 0 = 2010)	All documents excl. notifications	[run]	[run]	[run]	[run]	[run]
Armenia enters (2014)	Work permits	-54%	0.33	[run]	[run]	[run]
Armenia enters (2014)	All documents excl. notifications	x3.9	0.10	[run]	[run]	[run]
Armenia enters (2014), post-Soviet comparison origins	All documents excl. notifications	roughly x2 [run]	[run]	[run]	[run]	[run]
Russia excluded (2022)	Work permits	-85%	0.07	[run]	[run]	[run]
Russia excluded (2022)	All documents excl. notifications	-93%	0.03	[run]	[run]	[run]
Event-study coefficients by event year with placebo 2.5th/97.5th percentiles	all cases	[run]				

Notes: k = number of placebo estimates with an absolute value at least as large as the treated estimate; N = number of estimable placebo estimates. p = k/N is the convention used in the submitted version; p = (k+1)/(N+1) counts the treated unit. Georgia gained access in December 2010: results are shown with event year 0 = 2011 (first full year, Table 5) and 2010 (Figure 4 heading). The Armenian case is repeated with post-Soviet comparison origins only.

Source: Authors' estimates based on MRPiPS registers.

Status: draft.

Table A9. Interrupted time series with standard errors

Panel	Series and window	Level change March 2022	Slope change March 2022	Level change June 2025	Slope change June 2025
A	Work permits, Jan 2021-Nov 2025	-42%*** (s.e. [run])	-0.023*** ([run])	-28%*** ([run])	-0.007 ([run])
A	Employer declarations, Jan 2021-Nov 2025	-63%*** ([run])	-0.030*** ([run])	-18%*** ([run])	0.087*** ([run])
A	Seasonal work permits, Jan 2019-Nov 2025	-86%*** ([run])	0.002 ([run])	-19% ([run])	0.100*** ([run])
A	Notifications, Apr 2022-Nov 2025			3% ([run])	0.030*** ([run])
A	All documents excl. notifications, Jan 2021-Nov 2025	-59%*** ([run])	-0.024*** ([run])	-21%*** ([run])	0.049*** ([run])
A	All documents, Jan 2021-Nov 2025	-18%*** ([run])	-0.011* ([run])	-8%*** ([run])	0.044*** ([run])
B	Employer declarations, Jan 2015-Dec 2019 (break January 2018)	-39%*** ([run])	-0.023*** ([run])		
B	Work permits, half-yearly 2015H1-2021H2 (break 2018)	[run]	[run]		
A (extended)	Window to June 2026 if data are available	[run]	[run]	[run]	[run]
	Absolute post-break slopes, pulse and calendar-month effects	[run]			

Notes: Panel A: January 2021 - November 2025; Panel B: January 2015 - December 2019 (break January 2018) and half-yearly work permits 2015-2021 (time in years). OLS on log counts (log(1+y) for seasonal permits) with a linear trend and calendar-month effects; February 2022 pulse in Panel A. Newey-West standard errors (6 lags monthly, 2 lags half-yearly). Level changes in percent; slopes in log points per month. The June-November 2025 segment has six observations; slope changes are relative to the pre-break trend after calendar-month effects, so a positive slope change can coincide with a falling raw series.

Source: Authors' estimates based on MRPiPS registers.

Status: draft.

Table A10. Duration-sensitive concentration

Measure	2021	2022	2023	2024	2025
HHI, all documents excl. notifications, weighted by document-months	[run]	[run]	[run]	[run]	[run]
HHI, all documents, weighted by document-months	[run]	[run]	[run]	[run]	[run]
HHI, holders of a current work-authorising document (31 Dec 2025)					[run: requires holders by citizenship]
Effective number (1/HHI), same rows	[run]	[run]	[run]	[run]	[run]

Notes: Document-month weights use the maximum duration by channel and year (data/policy/document_duration.csv): permits 36, declarations 6 (24 from 2022), seasonal permits 9; notifications have no fixed maximum and the assumed weight is stated in that file. Holders: citizens holding a current work-authorising document on 31 December 2025.

Source: Authors' calculations based on MRPiPS registers and MRPiPS (2026).

Status: draft.

Table A11. Applications and issuance, 2025

Indicator	Group	Period	Value	Source
Work-permit applications	all foreigners	2025 H1	148,700	MRPiPS (2025b)
Work-permit applications	all foreigners	2025 H2	151,000	[confirm source]
Work permits issued	all foreigners	2025 H2	107,000	MRPiPS registers
Work permits issued, change on previous year	origins without declaration or notification access	January-May 2025	-11%	MRPiPS registers
Work permits issued, change on previous year	origins without declaration or notification access	June-December 2025	-31%	MRPiPS registers
Seasonal permits issued, change on previous year	origins without declaration or notification access	2025	[run] (fell)	MRPiPS registers
Pending cases	all foreigners	31 Dec 2024	58,000	MRPiPS (2025a)
Pending cases	all foreigners	31 Dec 2025	69,500	MRPiPS (2026)
Average processing time (days)	all foreigners	2024	64	MRPiPS (2025a)
Average processing time (days)	all foreigners	2025	82	MRPiPS (2026)
Applications and issuance by access group and half-year	both groups	2024-2025	[run]	MRPiPS registers

Notes: Applications, pending cases and issued permits need not refer to the same cohorts. 'Origins without declaration or notification access': all origins except ARM, BLR, GEO, MDA, RUS and UKR.

Source: MRPiPS (2025a, 2025b, 2026); authors' calculations based on MRPiPS registers.

Status: draft.

Table A12. Person-based benchmarks

Indicator	Ukrainian citizens	All foreigners	Ukrainian share	Reference date	Source
Holders of a current work-authorising document	582,507	1,241,232	47%	31 Dec 2025	MRPiPS (2026) [page]
Foreigners insured in ZUS	857,000	1,289,000	66.5%	31 Dec 2025	MRPiPS (2026), 15-20 (rounded counts)
Residence-and-work permits issued	251,000	328,000	77%	2025	MRPiPS (2026) [page]
Documents issued in the four channels	[run]	1,630,000	71%	2025	MRPiPS registers
Illustrative share incl. residence-and-work permits (no deduplication)			about 72%	2025	authors' calculation
Declarations per person		1.51 (2019; 1.50 with the rounded counts 1.64m/1.09m); 1.31 (2025)		2019; 2025	MRPiPS registers [enter unrounded counts]
Notifications filed vs active at year-end	about 1.03 million vs about 445000			2025	MRPiPS (2026) [page]
Declarations not taken up (employer reports)		32% / 39% / 27%		2023 / 2024 / 2025	MRPiPS (2025a, 2026)

Notes: The populations and reference periods differ: documents issued in a year, holders of a current document at year end, and persons insured in ZUS. None of these figures validates another directly; residence-and-work permits are not deduplicated against the four registers.

Source: MRPiPS (2025a, 2026); MRPiPS registers.

Status: draft.

Table A13. Variable definitions

file	name	label	unit	source
panel_origin_year.csv	iso3	Origin (citizenship) ISO 3166-1 alpha-3 code		crosswalk
panel_origin_year.csv	polish_name	Citizenship label in MRPiPS tables (Polish)		MRPiPS
panel_origin_year.csv	year	Calendar year	year	
panel_origin_year.csv	permits	Work permits issued (types A-E)	documents	MRPiPS
panel_origin_year.csv	declarations	Employer declarations registered (entered into the register from 2018)	documents	MRPiPS
panel_origin_year.csv	seasonal	Seasonal work permits issued	documents	MRPiPS
panel_origin_year.csv	notifications	Notifications of employment of Ukrainian citizens	documents	MRPiPS
panel_origin_year.csv	docs_excl_notif	All documents excluding notifications	documents	derived
panel_origin_year.csv	docs_all	All documents (four channels)	documents	derived
panel_origin_year.csv	channel_available_*	Channel in operation in the year		derived
halfyear_permits.csv	half	Half-year	1/2	MRPiPS
monthly_national.csv	ym	Month	YYYY-MM	MRPiPS
declaration_persons.csv	persons	Persons covered by declarations	persons	MRPiPS
zus_stock.csv	insured_foreigners	Foreigners insured in ZUS at year end	persons	MRPiPS reports
zus_stock.csv	insured_ukrainians	Ukrainian citizens insured in ZUS at year end	persons	MRPiPS reports
policy_coding.csv	decl_access_share	Share of the calendar year with access to employer declarations		legal acts (access_spells.csv)
policy_coding.csv	eu_visa_free_share	Share of the year with EU visa freedom (biometric passports)		European Commission (2026)
policy_coding.csv	decl_ever	Ever had declaration access (ARM BLR GEO MDA RUS UKR)		derived
policy_coding.csv	decl_group_2018	Six declaration origins x year >= 2018		derived
policy_coding.csv	decl_group_2022	Six declaration origins x year >= 2022		derived
policy_coding.csv	shock_ua2014	Ukraine x year >= 2014		derived
policy_coding.csv	shock_ua2022	Ukraine x year >= 2022		derived
policy_coding.csv	shock_by2021	Belarus x year >= 2021		derived
policy_coding.csv	shock_ru2022	Russia x year >= 2022		derived
panel_covariates.csv	flow_recent	ln(1 + documents in t-1 to t-3)	log documents	derived
panel_covariates.csv	flow_older	ln(1 + documents in t-4 to t-6)	log documents	derived
panel_covariates.csv	flow_recent_permits	ln(1 + permits in t-1 to t-3)	log documents	derived
panel_covariates.csv	flow_zero	Recent flow equals zero		derived
panel_covariates.csv	flow_x_access / flow_x_other	Recent flow x (non-)declaration-access group		derived
panel_covariates.csv	ln_presample_mean	ln mean permits 2008-2010 (0 if mean is 0)		derived
panel_covariates.csv	rel_income_l1	Relative real income RY(j t-1)	log points	World Bank NY.GDP.PCAP.KD.ZG
panel_covariates.csv	rel_income_l2	Relative real income lagged two years	log points	World Bank
panel_covariates.csv	ppp_gap_l1	ln(PPP GDP pc Poland / origin) t-1	log points	World Bank NY.GDP.PCAP.PP.CD
panel_covariates.csv	usd_gap_l1	ln(current-USD GDP pc Poland / origin) t-1	log points	World Bank NY.GDP.PCAP.CD
panel_covariates.csv	unemp_l1	Origin unemployment rate t-1 (ILO modelled)	percent	World Bank SL.UEM.TOTL.ZS
panel_covariates.csv	ln_pop_l1	ln origin population t-1	log persons	World Bank SP.POP.TOTL
panel_covariates.csv	ln_dist	ln distance Warsaw - origin capital	log km	GeoNames
panel_covariates.csv	contiguity	Shares a land border with Poland		crosswalk
panel_covariates.csv	former_ussr	Former Soviet republic		crosswalk
panel_covariates.csv	slavic_language	Slavic official language		crosswalk
regression_sample.csv	baseline	In the fixed-effects baseline sample (Table 3 col. 2)		derived

file	name	label	unit	source
regression_sample.csv	excl_UA_BY	Baseline without Ukraine and Belarus		derived
regression_sample.csv	never_eligible	Baseline origins never eligible for declarations		derived
regression_sample.csv	comparison70	Comparison origin for the event studies		derived

Notes: Full codebook: codebook/codebook.md and codebook/variables.csv.

Source: Replication package.

Status: draft.

Table A14. Software and runtime

Item	Value
Software used for the estimates in the article	[authors: e.g. Stata 18 MP; ppmlhdfe version; reghdfe; ftools - see reviewer comment 1]
Reference implementation in this package	Python 3.11+ (numpy; pandas; scipy; statsmodels; matplotlib) - see environment/software.md
Random seed	20261025
Pairs cluster bootstrap draws	999
Wild cluster score bootstrap draws	9999 (Rademacher and Webb weights)
Runtime of run_all	[run] (output/logs/runtime.csv)
Hardware	[authors]

Notes: Software versions are recorded at run time by code/10_appendix_tables.py; runtimes are taken from output/logs/runtime.csv.

Source: Replication package.

Status: draft.